

Roll No.

TCS-501

**B. TECH. (CSE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

SYSTEM SOFTWARE

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Providing the static address to the program during the assembling of the program is not best way in terms of multiprogramming. Suggest an alternative of this that will increase the efficiency of the multiprogramming system. Justify, how the suggested method is more efficient with example. Explain the suitable changes required in object program for the same. (CO2)

OR

- (b) Write the target addresses for the following instructions (OPCODE for STA is 0C (in Hex)) : (CO2)

OP	nixbpe	Disp/address	Target Address
000011	110010	0110 0000 0000	?
000011	111100	0011 0010 0001	?
000011	010000	0000 0011 0000	?
000011	110001	0000 1001 0010 0000 0010	?
000011	100010	0010 0000 1000	?

P. T. O.

Consider the following data :

(B) = 006000

(PC) = 003000

(X) = 000090

2. (a) Assembler needs to store the information related to the labels, operation codes and location. Identify the way assembler uses to store the details of the same. Explain with the help of block diagram how these methods contribute in the different passes of the assembler. (CO2)

OR

- (b) Explain the following machine independent features of SIC assembler : (CO2)

- (i) Symbol defining statements
- (ii) Expressions
- (iii) Control sections

3. (a) What is the difference between the sequences of the following statements ? (CO2)

(i) LDA #3

(ii) THREE EQU 3

Target Address	Display Address	opcode	Op
?	0110 0000 0000	110010	000011
?	0011 0010 0001	001111	000011
(iii) THREE	0000 0011 0000	010000	000011
?	0000 1001 0010 0010	110001	000011
?	0001 0000 0100	010001	110000
	LDA THREE		

OR

- (b) Assemble the following SIC/XE object code and show object code as would appear in object program : (CO2)

STCPY	START	1000
FIRST	LDX	ZERO
MOVECH	LDCH	STR1,X
	STCH	STR2,X
	TIX	ELEVEN
	JLT	MOVECH
STR1	BYTE	C'TEST STRING'
STR2	RESB	11
ZERO	WORD	0
ELEVEN	WORD	11
	END	FIRST

Assume the opcodes for mnemonics (LDX-04; LDCH - 50; STCH-54; TIX - 2C; JLT -38).

4. (a) Explain Expression, Literals and Literal Pools. Give the difference between immediate addressing and literals. (CO2)

OR

- (b) Calculate the target address and the value loaded in register A for the following machine instructions, consider the register and memory content as given in figure ahead : (CO2)

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(B) = 00A000

(PC) = 00B000

(X) = 000008

Address	Content
00A070	00A124
00A124	00CAB2
00B1A4	00A124

(i) 024070

(ii) 0121A4

(iii) 0321A4

(iv) 0310A070

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(CO3)

5. (a) Consider an SIC/XE program mentioned below :

Line	Loc	Source Statement
5	0000	COPY
10	0000	FIRST
		STL
		RETADR

40	0017	J	CLOOP
45	001A	ENDFIL	LDA
			=C 'EOF'

93		LTORG
	*	=C 'EOF'

135	1040	RLOOP	TD	INPUT
145	1046		RD	INPUT

185	105C	INPUT	BYTE	X 'Fi'
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215	1062	WLOOP	TD	=X '05'
230	106B		WD	=X '05'
255			END	FIRST
	1076			=X '05'

Suppose we made the following changes to the program :

- (i) Delete the LTORG statement on line 93.
- (ii) Change the statement on line 45 to + LDA.
- (iii) Change the operands on lines 135 and 145 to use literals (and delete the line 185).

Show the resulting object code for lines 45, 135, 145, 215 and 230. Also show the literal pool with address and data values.

The opcodes for the Mnemonics are as STL -14, J - 3C, LDA - 00, TD - E0, RD - D8, WD - DC.

OR

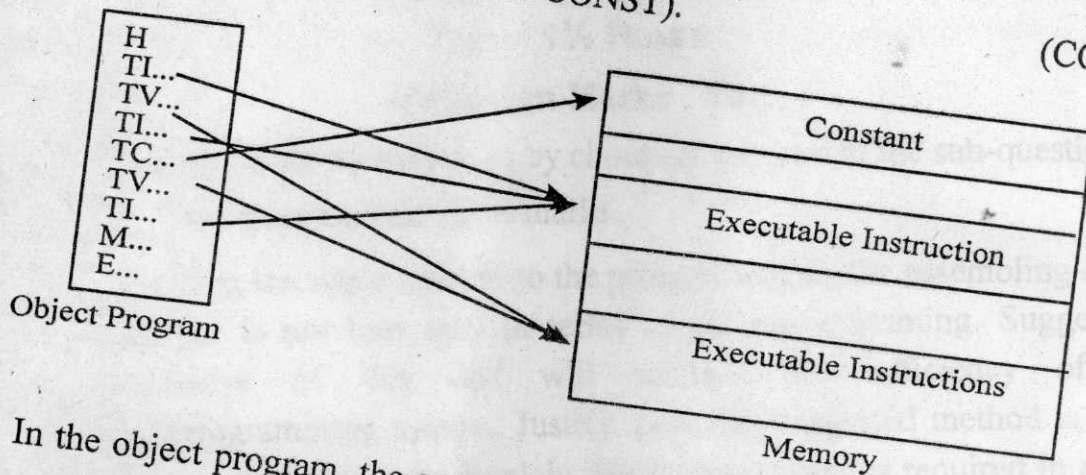
- (b) Suppose that a relocatable SIC/XE program is to be assembled in three different parts. One part contains the assembled instructions of the

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program (LDA, JSUB, etc.) Another part contains the data variables used in the program (which are defined by RESW, BYTE, RESB and WORD) : The third part contains data constants (which are defined by a new assembler directive named CONST).

(CO3)



In the object program, the assembled instructions are contained in type TI records, the variable in type TV records, and the constant in type TC records. (There new record types take the place of the normal text records in the object program.) The three parts or the object program will be loaded into separate areas of memory, as illustrated above.

Describe how the assembler could separate the object program TI, TV and TC records as described above. What will be advantages and disadvantages of such type of object program ?